Complete k2 row-combination synthesis

[pcc:unverified] DEC-20261102-592897 · TASK-20261102-b65f12 · 2026-11-05

[pcc:unverified] Decision: support the source-specific conditional algebraic theorem at analytical-review scope. PCC standing is unverified: no claim ID or signed verdict is supplied by this synthesis. Independent review TASK-20261102-bf48ac established all eight assigned joints. This is handwritten analytical evidence, not proof-assistant verification or a software experiment. Official archival completion remains subject to the exact report/decision archive.

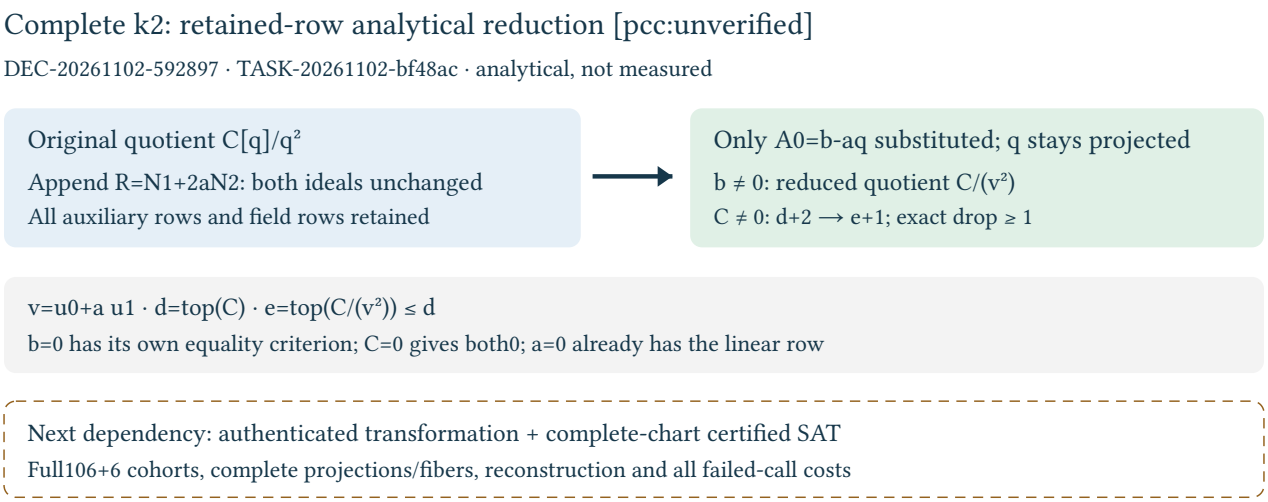

Figure 1: PCC-unverified analytical assertions; separate publication gate.

Observation: complete retained system

[pcc:unverified] For every prime p>3, nonsingular curve $y^2=X^3+\alpha X+\beta$ over F_p, finite target (a,b), valid interval 2^s<p and each of three rationality arms, the constant-B arity2 normalization gives A=A0+qX, B=1, $U=X^2+u1X+u0$. The norm is $A^2-f+(X-a)U$ and the sign row is L=A0+aq-b. All stratum, interval and actual imported rationality equations remain. The source binding loop makes them independent of A0 and q: the uniform arm binds only U; both stratified constant arms bind B0=1; every imported auxiliary is separately registered. Hidden component formulas and runtime imports are not revalidated by this textual separation proof.

[pcc:unverified] The analytical input explicitly appends x^p-x for every declared variable. This is a distinct presentation from canonical arithmetic output and from Boolean/range compilation. Let H contain all auxiliary rows and their field rows, K its highest-part ideal, and $C=F_p[z]/K$. Since every z_i^p belongs to K, C is finite, possibly zero. No ordinary-solution existence assumption is used.

Comparison: exact quotient and strict decrease

[pcc:unverified] The original highest-part ideal is $I=(A0+aq,q^2,K)$, giving $C[q]/q^2$. For a!=0, substituting A0=b-aq into only the literal original-X rows gives exactly the same quotient; fewer variables alone do not imply strictness.

[pcc:unverified] Append $R=N1+2aN2=2q(A0+aq)+u0+a\,u1-\alpha-2a^2$ without deleting any old nonzero row. Its ordinary identity and its separate highest-part certificate R_top=2q L_top preserve both pre-pivot ideals. After the sole A0 pivot the retained added row becomes $2bq+v-\alpha-2a^2$, where v=u0+a u1. The complete reduced highest ideal is $J=(q^2,K,v+2bq)$.

[pcc:unverified] If C!=0 and b!=0, put d=top(C), e=top(C/(v²)). Explicit inverse graded maps give S/J=C/(v²), with e<=d. Original regularity is d+2; reduced regularity is e+1. The exact drop is d-e+1>=1. No regular-element premise is required and no diagnostic value such as 2p is copied into the complete chart. These are derived algebraic values; no runtime samples or statistical uncertainty apply.

[pcc:unverified] If b=0, the quotient is $(C/(v))[q]/q^2$, with regularity e+2. Equality holds exactly when (C/(v))_d!=0, equivalently C_d != v C_(d-1); otherwise the decrease is strict. If C=0 both values are0. At a=0, R duplicates N1 and the raw list already contains its linear row. Zero/linear B strata contain literal constants and remain unit cases. Identity targets, special arities and other arities do not inherit this theorem.

Fidelity: q remains projected

[pcc:unverified] Over every extension of F_p, projection (A0,q,z)->(q,z) and reconstruction (q,z)->(b-aq,q,z) are inverse solution maps and ordinary coordinate-ring isomorphisms. Every corresponding complete z fiber is identical. Original projected pairs are exactly {(b-aq,q): q in the reduced projection}. The graded elimination of q used above is a proof calculation only: q remains a runtime projected input.

[pcc:unverified] The eliminated field row transforms to -a(q^p-q); all surviving variable field rows remain. Reconstructed A=b+q(X-a), B=1 and U, plus unchanged auxiliaries, restore the original signed-recovery inputs. Constant B=1 has no common A/B root. Both common-root beta-sign controls remain in their broader chart family; neither sign is silently selected. Bit-layout authentication, complete bit models and common signed-output comparison remain software obligations.

Controls and eight-joint reconciliation

[pcc:unverified] The reviewer established source_separation (D2), original_quotient (D3), raw_equality (D4), added_row (D5), reduced_quotient (D6), strictness_boundaries (D7-D8), solutions_projection (D9), and claim_boundary (D1,D4,D10). Coordinator read all D1-D11 and reconciled load-bearing source paths.

[pcc:unverified] Algebraic controls distinguish equality and strictness at b=0: C=F_p[u0,u1]/(u0+a u1,u1^p) gives v=0 and equal p+1; C=F_p[u0,u1]/(u1,u0^p) gives values p+1 and2. The first ring gives a sharp drop exactly1 at nonzero b. These rings are not asserted to be realizations of actual chart auxiliaries. Missing field rows can invalidate finiteness; coefficients outside F_p break the field-row certificate; ordinary-redundant row deletion can alter highest-part regularity; odd-characteristic Boolean idempotence requires an extra cross term after x=y+z. A q-dependent residual defeats separation. Ordinary unsatisfiability need not make the highest-part quotient zero. These controls remain part of the supported result.

Inference and limitations

[pcc:unverified] The prospective prior that this certified row exposes lower-degree information survives independent challenge at the stated complete-k2 scope. Source-specific strict regularity reduction is supported with its nonzero-C and nonzero-ordinate premises. Solving degree, matrix work, certified pruning and SAT costs are separate quantities. No implementation, census, runtime consequence, cross-family equivalence, k3 extension or duplicate-inverse draft is accepted by this decision.

Custody and procedure

[pcc:unverified] At the Coordinator pre-merge readback on 2026-11-05, before the 15:10:07Z ledger epoch2 acquisition, the four sealed review artifacts and all13 frozen inputs matched their recorded hashes. After the prospective policy transition, AGENTS.md is historically bound to source commit3a4647618e91b2a4fb6f2b32caf923a25ce27e39; the12 remaining frozen source inputs are unchanged. This is a pre-merge readback with a historical procedural-input binding, not an assertion that current AGENTS.md retains its old bytes. The native independence gate returned0 for the frozen eight-joint plan; this checks declared source/ownership consistency, not mathematical truth or model identity. Review dispatch was2026-11-03T18:04:31.999139Z; sealing was18:21:06.942099Z, elapsed994.94296s within the original3600s deadline. One session, one round, no extension or blindness lift. Scientific runs, imports, suites, CAS, solvers, censuses and remote jobs were all zero; peak memory is not reported as measured.

[pcc:unverified] The exact Caminata-Gorla v2 PDF https://arxiv.org/pdf/2112.05579v2 was retrieved and read by TASK-20261102-bf48ac before preliminary algebra. Definitions1.1 and1.4 on PDF pages4 and6 distinguish solving degree from highest-part regularity. PDF SHA-256: 9e9323db0cd2f755a375c1baa836b999972fc38eebca3e6f55f63dfecb226177. It supplies definitions only. Coordinator relies on that retrieved-source attestation and did not refetch the paper.

[pcc:unverified] Requested review policy was review-adversarial/xhigh. Actual provider/model/tier/effective effort and native capacities remain unauthenticated under the approved prospective amendment. Root reports review publication90606c009d6297c5bc2166681b28bae78dd6a183 and release55b2dac99b. Ledger epoch1 expired without official outputs; epoch2 was acquired15:10:07Z Nov5, expires17:10:07Z, published5a3c0ff471. Root owns claims and Git.

[pcc:unverified] Coordinator-only read-scope deviation: ledger/decisions/DEC-20261102-cd6ac3.yaml and docs/reports/prime-sat-affine-regularity-20261102/report.typ were read for prior record/rendering conventions; .resume/complete-k2-row-combination-review-root-verification.json and .resume/complete-k2-row-combination-review-independence-gate.json for root custody/gate context. These were outside the handoff input list and are disclosed, not retrospectively authorized. They did not change the already-sealed reviewer blindness. Current mathematical support rests on allowed independent proof and source. Procedural coordinate/research-visuals skills were also read; no further unlisted artifacts were accessed.

Retained scope and next action

[pcc:unverified] Keep all106 coordinate cells over F17(2,2), F97(1,4), widths2/3, all affine-x least-positive-y targets, quartic/membership/quadratic/quadratic_aa; separately keep all six older F17 coefficient cells, degrees8/16 and three targets, leaf_sat/prefix_univariate_gcd/exhaustive_reference/aa_reverse_prefix. Keep all15 chart fixtures across three arms/public strata, eight nilpotent and eight guard fixtures, mutation/certificate controls, both common-root beta signs, wrapped intervals and targets outside interval. All five leads remain; unchanged partial-matrix/two-bit-backdoor scaling is not admitted.

[pcc:unverified] Charge construction, row addition/identity checks, substitution, field rows, repeated authentication, compilation, every attempted solver call, certificates, failed/censored work, decoding, reconstruction, signed checking, full projection and blocking. Complete model sets and corresponding fibers must survive; small fixtures cannot replace either original cohort.

[pcc:unverified] Next, prepare the authenticated retained-row transformation and complete-chart certified SAT integration, including complete failure accounting and narrowly versioned census zero semantics. Prioritize this implementation dependency over prospective k3 theory. No implementation or experiment is admitted here. The latest owner cloud limits supersede older advisory/unlimited guidance: local CPU/RAM/time/attempt caps alone do not supply a finite dollar allocation or live remaining-budget evidence. Cloud build/validation/experiment admission still requires those limits, lawful live resource checks and exact GitHub publication.

[pcc:unverified] The scoped theorem warrants knowledge curation after archive; no knowledge ID or write path exists in this handoff. Preserve the analytical finding and open software/pruning question for that action. The inspected producer diagram remains an immutable UNREVIEWED snapshot; the new decision diagram is the current scoped analytical view. No broader frontier bound is changed. Root renders/inspects the PDF, binds the input manifest and publishes exact review4 plus ledger6; no goal/hypothesis/experiment status is changed.

Evidence bindings

[pcc:unverified] Internal provenance: independent-derivation.md, source-map.json, review.md and attestation.yaml under research/prime_sat_complete_k2_row_combination_20261102/TASK-20261102-bf48ac/. Full hashes and all13 input bindings are in DEC-20261102-592897.yaml.

[pcc:unverified] attestation.yaml: 669039e333e39278dc56d9bba419a8f4ad524171144182c056f951f45c322765
independent-derivation.md: 5cc5e12088b8c05f3ca6a6aca5fa1d56b21fc0565e0bcfd0aa30bb4439b75bc7
review.md: c349db5ceb52a59783862ca632eacf60a654afda8162ee093d2379c837df70bc
source-map.json: 3058e2762bb92a4a0ed9b0850115b49a30deb96870a15405b195f46e2c1efed2

Prospective publication-policy transition

[pcc:unverified] After these four draft sources were written and before sealing, root reported fetched main6e682726fa changes AGENTS.md and introduces docs/pcc.md. Root authorized the specific policy read via git show origin/main:docs/pcc.md. All result assertions here are PCC-unverified; analytical review support is not signed standing. Separate PCC claim creation and policy/custody reconciliation remain publication dependencies outside this handoff write scope. No sealed review history is rewritten. The13 matching input hashes describe the Coordinator pre-merge readback, not changed procedural bytes after integration. Root authorized finalizing analytical sources with PCC-unverified labels while historical/prospective policy custody and publication remain root responsibilities.

Minimal PCC publication supplement

[pcc:unverified] The Coordinator authorizes one native theorem claim, pinning the sealed independent derivation and review as unsigned attestation subjects, plus exactly its native append-only ledger claim entry. The actual generated claim path, ID, hash and standing must be captured by root. No signature, key, keyring change, Lean proof, new experiment or research ID is authorized. These files belong to a separate supplemental publication commit; the exact10-path analytical archive remains intact. The theorem statement retains all p>3, finite-k2, constant-B, full-field, nonzero-C and nonzero-b premises, all retained rows, field-solution/fiber reconstruction and projected q. Native theorem verification requires Lean evidence, which is absent; document pins do not supply it. Root may run read-only standing, ledger-prefix and prose checks and must inspect uncertain outcomes before retrying the one creation.

[pcc:unverified] Prospective instruction context additionally read with explicit root authorization: fetched tools/pcc/pcc.py CLI/schema and docs/SABOTAGE_RESISTANCE.md. This authorization is separate from the disclosed earlier Coordinator read-scope deviation.